Circles

Class 4

Note for the teacher.

- **Aims.** Students decide when four points lie on one circle. They explain why a segment that subtends equal angles at two points on the same side gives four concyclic points, why opposite angles of a cyclic quadrilateral add up to 180°, and the converse. They use these results to find angles and to recognise cyclic quadrilaterals.
- **NCERT links.** Theorems 10, 11 and 12 keep their NCERT numbers. The questions include the NCERT in-text exercise in Section 5.8 and End-of-Chapter Questions 7, 8, 10, 14, 15 and 21.
- **Prior work.** Class 3: Theorem 9, angles in the same segment, the angle in a semicircle, and P7 (a point inside the circle gives a larger angle, a point outside a smaller one).
- **Materials.** One A4 sheet and the geometry box for each student.
- **Method.** Theorems 10 and 12 are proved by contradiction. P3 and P6 lead students through each argument one step at a time; let them write each step before it is discussed.

1 When do four points lie on a circle?

Points that lie on one circle are called **concyclic**. Any three points that are not on one line are concyclic (Theorem 5). What about four points?

Solve first.

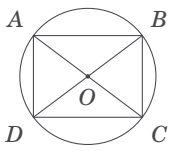
P1. (a) $ABCD$ is a rectangle with diagonals meeting at O. Draw a rough figure of the rectangle and its diagonals. Explain why $OA = OB = OC = OD$. Are A, B, C and D concyclic?

(b) On an A4 sheet, draw $PQ = 4$ cm. At P, draw $\angle QPS = 60°$ with $PS = 4$ cm. With centres Q and S and radius 4 cm, draw arcs that meet at R. Join QR and SR; $PQRS$ is a rhombus. Now draw the circle through P, Q and S: construct the perpendicular bisectors of PQ and PS, mark their meeting point O, and draw the circle with centre O and radius OP. Does this circle pass through R?

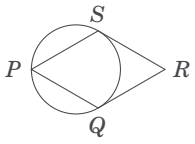
Answers.

P1.

(a) The diagonals of a rectangle are equal and bisect each other, so $OA = OB = OC = OD$. The circle with centre O and radius OA passes through all four vertices: they are concyclic.



(b) No. $\triangle PQS$ is equilateral, and its circumcircle has radius about 2.3 cm. R lies outside this circle. So four points need not be concyclic.



2 Equal angles on the same side

Solve first.

P2. A, B, C are points on a circle. D is a point on the same side of AB as C, and $\angle ADB = \angle ACB$. Suppose D is *outside* the circle. Then AD cuts the circle at a point E between A and D. Draw a rough figure: the circle through A, B and C, the point D outside it, and the segments CA, CB, DA, DB and EB.

(a) Why is $\angle AEB = \angle ACB$?

(b) $\angle AEB$ is an exterior angle of $\triangle BED$. Which is larger, $\angle AEB$ or $\angle ADB$?

(c) Use (a), (b) and $\angle ADB = \angle ACB$. What goes wrong?

(d) What goes wrong if D is *inside* the circle? (Extend AD to meet the circle at E.)

(e) So where must D be?

Answers.

P2.

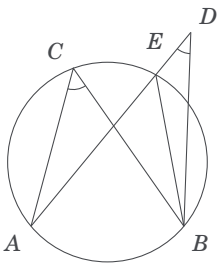
(a) C and E are in the same segment of chord AB, so the angles are equal (Class 3).

(b) An exterior angle is greater than each interior opposite angle, so $\angle AEB > \angle ADB$.

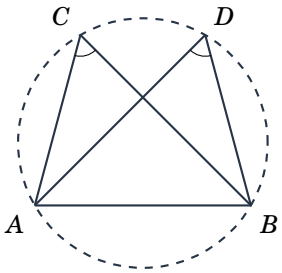
(c) Then $\angle ACB = \angle AEB > \angle ADB = \angle ACB$: an angle greater than itself. This is impossible.

(d) Now $\angle ADB$ is the exterior angle of $\triangle BDE$, so $\angle ADB > \angle AEB = \angle ACB$. This is also impossible.

(e) On the circle through A, B and C.



Theorem 10 (NCERT Theorem 10). *If a line segment AB joining two points A, B subtends equal angles at two other points C, D that lie on the same side of AB, then the four points lie on a circle.*



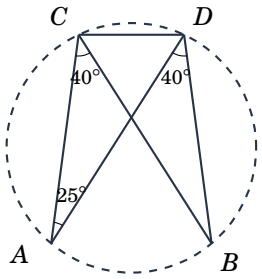
Why it works.

Given: C and D lie on the same side of AB, and $\angle ACB = \angle ADB$.
To show: A, B, C and D lie on one circle.

1. A, B and C are not on one line, so one circle passes through them (Theorem 5).
2. If D were outside this circle, P2(c) shows that $\angle ACB > \angle ACB$. If D were inside, P2(d) gives the same kind of contradiction.
3. So D lies on the circle.

Theorem 10 is the converse of "angles in the same segment are equal".

Example 1. C and D lie on the same side of AB, with $\angle ACB = \angle ADB = 40°$ and $\angle CAD = 25°$. Find $\angle CBD$.



Given: $\angle ACB = \angle ADB = 40°$, C and D on the same side of AB, $\angle CAD = 25°$.
To find: $\angle CBD$.

1. A, B, C and D are concyclic (Theorem 10).
2. $\angle CBD$ and $\angle CAD$ stand on the chord CD, in the same segment. So $\angle CBD = \angle CAD = 25°$.

Answer: $\angle CBD = 25°$.

3 Cyclic quadrilaterals

A quadrilateral whose four vertices lie on a circle is a **cyclic quadrilateral**.

Solve first.

P3. $ABCD$ is a cyclic quadrilateral in a circle with centre O. The arc BCD subtends $\angle BOD = 140°$ at O. Draw a rough figure: the circle, its centre O, the quadrilateral $ABCD$, and the radii OB and OD.

(a) A is outside the arc BCD. Find $\angle BAD$.
(b) What angle does the arc BAD subtend at O? Find $\angle BCD$.
(c) Find $\angle BAD + \angle BCD$.

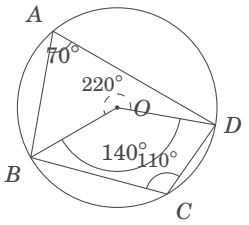
P4. Repeat P3 with $\angle BOD = 100°$. What do you notice?

P5. In P3, find $\angle ABC + \angle ADC$ without any more information.

Answers.

P3.

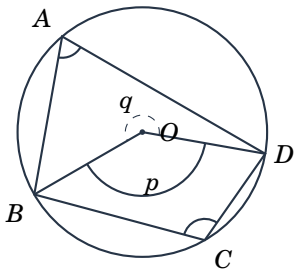
(a) $\angle BAD = \dfrac{140°}{2} = 70°$ (Theorem 9).
(b) The reflex angle $360° - 140° = 220°$. C is outside the arc BAD, so $\angle BCD = \dfrac{220°}{2} = 110°$.
(c) $70° + 110° = 180°$.



P4. $\angle BAD = 50°$ and $\angle BCD = \dfrac{260°}{2} = 130°$. The sum is again 180°: the two arcs together make the whole circle, 360°, and each angle is half of its arc's angle.

P5. The angles of a quadrilateral add up to 360°. So $\angle ABC + \angle ADC = 360° - 180° = 180°$.

Theorem 11 (NCERT Theorem 11). *The sum of two opposite angles of a cyclic quadrilateral is* 180°.



Why it works.

Given: $ABCD$ is a cyclic quadrilateral in a circle with centre O.
To show: $\angle BAD + \angle BCD = 180°$.

1. The arc BCD subtends an angle p at O, and the arc BAD subtends the rest of the complete angle, $q = 360° - p$.
2. A is outside the arc BCD, so $\angle BAD = \dfrac{p}{2}$. C is outside the arc BAD, so $\angle BCD = \dfrac{q}{2}$ (Theorem 9).
3. $\angle BAD + \angle BCD = \dfrac{p+q}{2} = \dfrac{360°}{2} = 180°$.

The other pair also adds up to 180°, because the four angles add up to 360° (P5).

Solve first.

P6. $ABCD$ is a quadrilateral with $\angle BAD + \angle BCD = 180°$. Suppose C is *outside* the circle through A, B and D, and CD cuts this circle at E, between D and C. Draw a rough figure: the circle through A, B and D, the point C outside it, the quadrilateral $ABCD$, and the segment BE.

(a) $ABED$ is a cyclic quadrilateral. Find $\angle BAD + \angle BED$.

(b) Compare $\angle BED$ with $\angle BCD$.

(c) $\angle BED$ is an exterior angle of $\triangle BEC$. What goes wrong?

(d) So where must C be?

Answers.

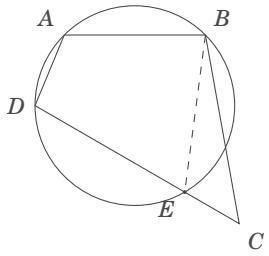
P6.

(a) 180° (Theorem 11).

(b) Both $\angle BED$ and $\angle BCD$ make 180° with $\angle BAD$, so $\angle BED = \angle BCD$.

(c) An exterior angle is greater than each interior opposite angle, so $\angle BED > \angle BCE = \angle BCD$. This contradicts (b).

(d) C cannot be outside the circle. Exercise 4A Question 10 shows that it cannot be inside either. So C is on the circle.



Theorem 12 (NCERT Theorem 12). *If two opposite angles of a quadrilateral add up to* 180°*, then the vertices of the quadrilateral lie on a circle, that is, they are concyclic.*

Why it works.

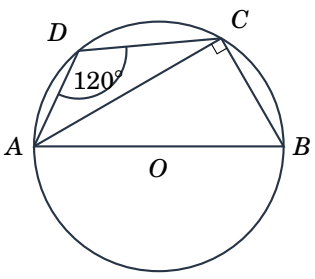
Given: quadrilateral $ABCD$ with $\angle BAD + \angle BCD = 180°$.
To show: $ABCD$ is cyclic.

1. Draw the circle through A, B and D (Theorem 5).
2. If C were outside it, P6 gives $\angle BED = \angle BCD$ and $\angle BED > \angle BCD$ together, which is impossible. The case of C inside is similar.
3. So C lies on the circle, and $ABCD$ is cyclic.

Things to Remember

Opposite angles of a cyclic quadrilateral add up to 180°. If opposite angles of a quadrilateral add up to 180°, it is cyclic.
If a segment subtends equal angles at two points on the same side of it, the four points are concyclic.

Example 2. $ABCD$ is a cyclic quadrilateral. AB is a diameter of the circle, and $\angle ADC = 120°$. Find $\angle BAC$.



Given: $ABCD$ cyclic, AB a diameter, $\angle ADC = 120°$.
To find: $\angle BAC$.

1. $\angle ABC = 180° - 120° = 60°$ (opposite angles of a cyclic quadrilateral, Theorem 11).

2. $\angle ACB = 90°$ (angle in a semicircle).
3. In $\triangle ABC$, $\angle BAC = 180° - 90° - 60° = 30°$.

Answer: $\angle BAC = 30°$.

For every question in Exercise 4A, draw a rough figure: the circle (where there is one), the quadrilateral or triangle named in the question, and every given angle and length marked on it.

Exercise 4A

1. $ABCD$ is a cyclic quadrilateral. If $\angle A = 75°$, find $\angle C$. If $\angle B = 110°$, find $\angle D$.

   Solution.

   Given: $ABCD$ is a cyclic quadrilateral, $\angle A = 75°$ and $\angle B = 110°$.
   To find: $\angle C$ and $\angle D$.

   1. $\angle C = 180° - 75° = 105°$ (opposite angles of a cyclic quadrilateral, Theorem 11).
   2. $\angle D = 180° - 110° = 70°$ (opposite angles of a cyclic quadrilateral, Theorem 11).

   Answer: $\angle C = 105°$, $\angle D = 70°$.

2. $PQRS$ is a cyclic quadrilateral with $\angle P = (2x + 10)°$ and $\angle R = (3x - 20)°$. Find x, $\angle P$ and $\angle R$.

   Solution.

   Given: $PQRS$ is a cyclic quadrilateral, $\angle P = (2x + 10)°$ and $\angle R = (3x - 20)°$.
   To find: x, $\angle P$ and $\angle R$.

   1. P and R are opposite angles, so $(2x + 10) + (3x - 20) = 180$ (Theorem 11).
   2. $5x - 10 = 180$, so $x = 38$ (solving the equation).
   3. $\angle P = 2(38) + 10 = 86°$ (substituting $x = 38$).
   4. $\angle R = 3(38) - 20 = 94°$ (substituting $x = 38$).

   Answer: $x = 38$, $\angle P = 86°$, $\angle R = 94°$.

3. A quadrilateral $ABCD$ has $\angle A = 80°$, $\angle B = 110°$, $\angle C = 100°$ and $\angle D = 70°$. Can it be cyclic? Explain.

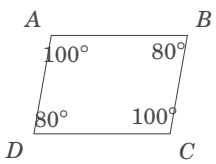
   Solution.

   Yes. $\angle A + \angle C = 80° + 100° = 180°$ and $\angle B + \angle D = 110° + 70° = 180°$, so $ABCD$ is cyclic (Theorem 12).

4. Shreya says, "Quadrilateral $ABCD$ has $\angle A = 100°$, $\angle B = 80°$, $\angle C = 100°$ and $\angle D = 80°$. Its angles $100°$ and $80°$ add up to $180°$, so it is cyclic." Find her mistake.

   Solution.

   Shreya is wrong. $100°$ and $80°$ are *adjacent* angles ($\angle A$ and $\angle B$), not opposite angles. The opposite angles are $\angle A$ and $\angle C$: $100° + 100° = 200° \neq 180°$, so $ABCD$ is not cyclic (Theorem 12); it is a parallelogram that is not a rectangle.

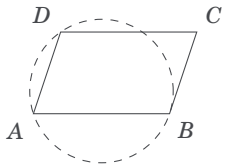
5. Four trees P, Q, R and S stand at the corners of a quadrilateral plot, with $\angle P = 75°$, $\angle Q = 95°$ and $\angle R = 105°$. A gardener wants to lay a circular path through all four trees. Is it possible? Explain.

   Solution.

   Yes. $\angle S = 360° - 75° - 95° - 105° = 85°$ and $\angle P + \angle R = 75° + 105° = 180°$, so the plot is cyclic (Theorem 12): one circular path passes through all four trees.

6. Show that a rectangle is the only parallelogram that can be cyclic.

Solution.

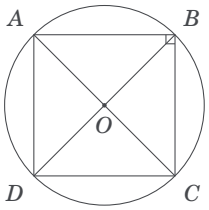


Given: $ABCD$ is a parallelogram and it is cyclic.
To prove: $ABCD$ is a rectangle.
Proof:

1. Opposite angles of a parallelogram are equal: $\angle A = \angle C$ (opposite angles of a parallelogram).
2. $\angle A + \angle C = 180°$ (Theorem 11).
3. So $2\angle A = 180°$, that is, $\angle A = \angle C = 90°$ (from the two steps above).
4. A parallelogram with a right angle is a rectangle (definition of a rectangle).
5. So the only cyclic parallelograms are rectangles, and a rhombus that is not a square is not cyclic (from the steps above).
6. Hence proved.

7. A rectangle is inscribed in a circle. Show that its diagonals meet at the centre of the circle.

Solution.

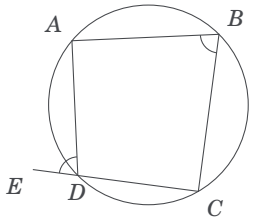


Given: rectangle $ABCD$ is inscribed in a circle with centre O.
To prove: the diagonals of $ABCD$ meet at O.
Proof:

1. $\angle ABC = 90°$ (angle of a rectangle).
2. The arc AC not containing B subtends $\angle ABC = 90°$ at B, so it subtends $2 \times 90° = 180°$ at the centre O (Theorem 9).
3. So A, O and C are on one line: AC is a diameter, and O is its midpoint (straight angle at O).
4. In the same way, BD is a diameter with midpoint O (same reasoning).
5. So the diagonals meet at O (both pass through O).
6. Hence proved.

8. $ABCD$ is a cyclic quadrilateral, and E is a point on CD extended beyond D. Explain why $\angle ADE = \angle ABC$.

Solution.

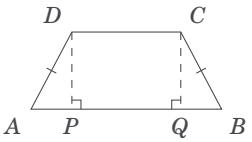


Given: $ABCD$ is a cyclic quadrilateral and E is a point on CD extended beyond D.
To prove: $\angle ADE = \angle ABC$.
Proof:

1. $\angle ADE + \angle ADC = 180°$ (linear pair).
2. $\angle ABC + \angle ADC = 180°$ (Theorem 11).
3. So the exterior angle equals the interior opposite angle, $\angle ADE = \angle ABC$ (subtracting $\angle ADC$ from the equal sums).
4. Hence proved.

9. $ABCD$ is a trapezium with $AB \parallel DC$ and $AD = BC$. Show that $ABCD$ is cyclic.

Solution.

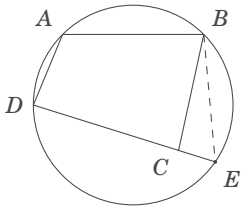


Given: $ABCD$ is a trapezium with $AB \parallel DC$ and $AD = BC$.
To prove: $ABCD$ is cyclic.
Construction: Draw $DP \perp AB$ and $CQ \perp AB$.
Proof:

1. In $\triangle APD$ and $\triangle BQC$, $AD = BC$ (given) and $DP = CQ$ (distance between the parallel lines), so $\triangle APD \cong \triangle BQC$ (RHS).
2. So $\angle A = \angle B$ (CPCT).
3. $AB \parallel DC$, so $\angle A + \angle D = 180°$ (co-interior angles).
4. So $\angle B + \angle D = 180°$, and $ABCD$ is cyclic (Theorem 12).
5. Hence proved.

10. In P6, suppose C is *inside* the circle through A, B and D. Extend DC to meet the circle at E. Show that this is also impossible.

Solution.



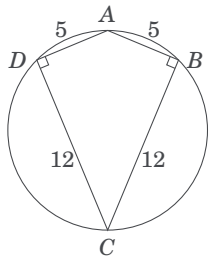
Given: $ABCD$ is a quadrilateral with $\angle BAD + \angle BCD = 180°$, and C lies inside the circle through A, B and D.
To prove: this is impossible.
Construction: Extend DC to meet the circle at E. Join BE.
Proof:

1. $ABED$ is cyclic, so $\angle BAD + \angle BED = 180°$ (Theorem 11).
2. $\angle BAD + \angle BCD = 180°$ (given), so $\angle BED = \angle BCD$ (both make $180°$ with $\angle BAD$).
3. C is between D and E, so $\angle BCD$ is an exterior angle of $\triangle BCE$; hence $\angle BCD > \angle BEC = \angle BED$ (exterior angle of a triangle).
4. The two statements contradict each other, so C cannot be inside the circle (contradiction).
5. Hence proved.

11. A cyclic quadrilateral has sides 5, 5, 12 and 12 units. Find its area.

Solution.

Given: a cyclic quadrilateral has sides 5, 5, 12 and 12 units.
To find: its area.



1. If the sides are in the order 5, 12, 5, 12, opposite sides are equal, so it is a parallelogram (opposite sides equal).
2. A cyclic parallelogram is a rectangle (Question 6).
3. Area $= 5 \times 12 = 60$ square units (area of a rectangle).
4. If the order is 5, 5, 12, 12, it is a kite $ABCD$ with $AB = AD = 5$ and $CB = CD = 12$ (two pairs of adjacent equal sides).
5. $\triangle ABC \cong \triangle ADC$ (SSS), so $\angle B = \angle D$ (CPCT).
6. $\angle B + \angle D = 180°$ (Theorem 11), and $\angle B = \angle D$, so each is $90°$ (halves of $180°$).
7. Area $= 2 \times \frac{1}{2} \times 5 \times 12 = 60$ square units (sum of the areas of the two right triangles).

Answer: 60 square units in both cases.

12. On an A4 sheet, draw a circle of radius 4 cm. Mark four points A, B, C and D on the circle, in order around it, and join AB, BC, CD and DA. Measure the four angles of $ABCD$ with a protractor, and add each pair of opposite angles. Does your result agree with Theorem 11? Then keep A, B and C, and mark a point S inside the circle, between the chord AC and the point D. Join AB, BC, CS and SA. Measure the angles of $ABCS$ and add each pair of opposite angles. Do they add up to $180°$?

Solution.

Steps of construction:

1. Draw a circle of radius 4 cm with centre O (using a compass).
2. Mark four points A, B, C, D on the circle, in order.
3. Join AB, BC, CD and DA to get quadrilateral $ABCD$.
4. Measure $\angle A$, $\angle B$, $\angle C$ and $\angle D$ with a protractor, and find $\angle A + \angle C$ and $\angle B + \angle D$.
5. Mark S inside the circle, between the chord AC and D, and join CS and SA to get quadrilateral $ABCS$.
6. Measure its four angles, and find $\angle A + \angle C$ and $\angle B + \angle S$.

Answer: For the cyclic quadrilateral, each pair of opposite angles adds up to about $180°$ (within $2°$ for measuring); for $ABCS$, $\angle B + \angle S$ is more than $180°$, because S is inside the circle (Class 3, P7).

Note for the teacher. Question 4 must be discussed in class: Shreya added two adjacent angles, not opposite ones. Questions 6, 7 and 9 together show which special quadrilaterals are cyclic. If time is short, set Questions 10, 11 and 12 as homework.

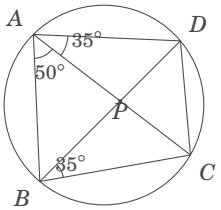
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4 Further questions

F1. $ABCD$ is a cyclic quadrilateral whose diagonals meet at P. $\angle DBC = 35°$ and $\angle BAC = 50°$. Find $\angle BAD$ and $\angle BCD$.

Solution.

Given: $ABCD$ is a cyclic quadrilateral whose diagonals meet at P, $\angle DBC = 35°$ and $\angle BAC = 50°$.
To find: $\angle BAD$ and $\angle BCD$.



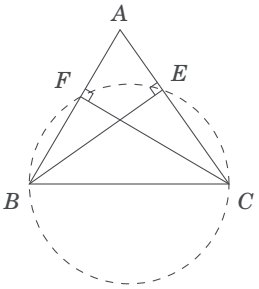
1. $\angle DAC = \angle DBC = 35°$ (angles in the same segment, chord DC).
2. $\angle BAD = \angle BAC + \angle CAD = 50° + 35° = 85°$ (the whole angle is the sum of its parts).
3. $\angle BCD = 180° - 85° = 95°$ (Theorem 11).

Answer: $\angle BAD = 85°$, $\angle BCD = 95°$.

F2. ABC is a triangle, and BE and CF are its altitudes. Show that B, C, E and F are concyclic.

Solution.

Given: ABC is a triangle, and BE and CF are its altitudes.
To prove: B, C, E and F are concyclic.
Proof:



1. $BE \perp AC$, so $\angle BEC = 90°$ (altitude).
2. $CF \perp AB$, so $\angle BFC = 90°$ (altitude).
3. So $\angle BEC = \angle BFC$ (both are 90°).
4. E and F lie on the same side of BC, so BC subtends equal angles at E and F, and B, C, E, F are concyclic (Theorem 10).
5. The circle has BC as a diameter (converse of the angle in a semicircle).
6. Hence proved.

5 The twelve theorems of the chapter

Class	Our no.	NCERT no.	Statement
1	1	2	Equal chords subtend equal angles at the centre.
1	2	3	Chords that subtend equal angles at the centre are equal.
1	3	4	The line from the centre to the midpoint of a chord is perpendicular to the chord.
1	4	5	The perpendicular from the centre to a chord bisects the chord.
2	5	1	There is a unique circle through three non-collinear points.
2	6	6	Equal chords are equidistant from the centre.
2	7	7	Chords equidistant from the centre are equal.
2	8	8	Of two chords, the longer one is nearer the centre.
3	9	9	The angle at the centre is double the angle at a point on the rest of the circle. *Corollary:* the angle in a semicircle is $90°$.
4	10	10	If AB subtends equal angles at C and D on the same side, then A, B, C, D are concyclic.
4	11	11	Opposite angles of a cyclic quadrilateral add up to $180°$.
4	12	12	If opposite angles of a quadrilateral add up to $180°$, it is cyclic.